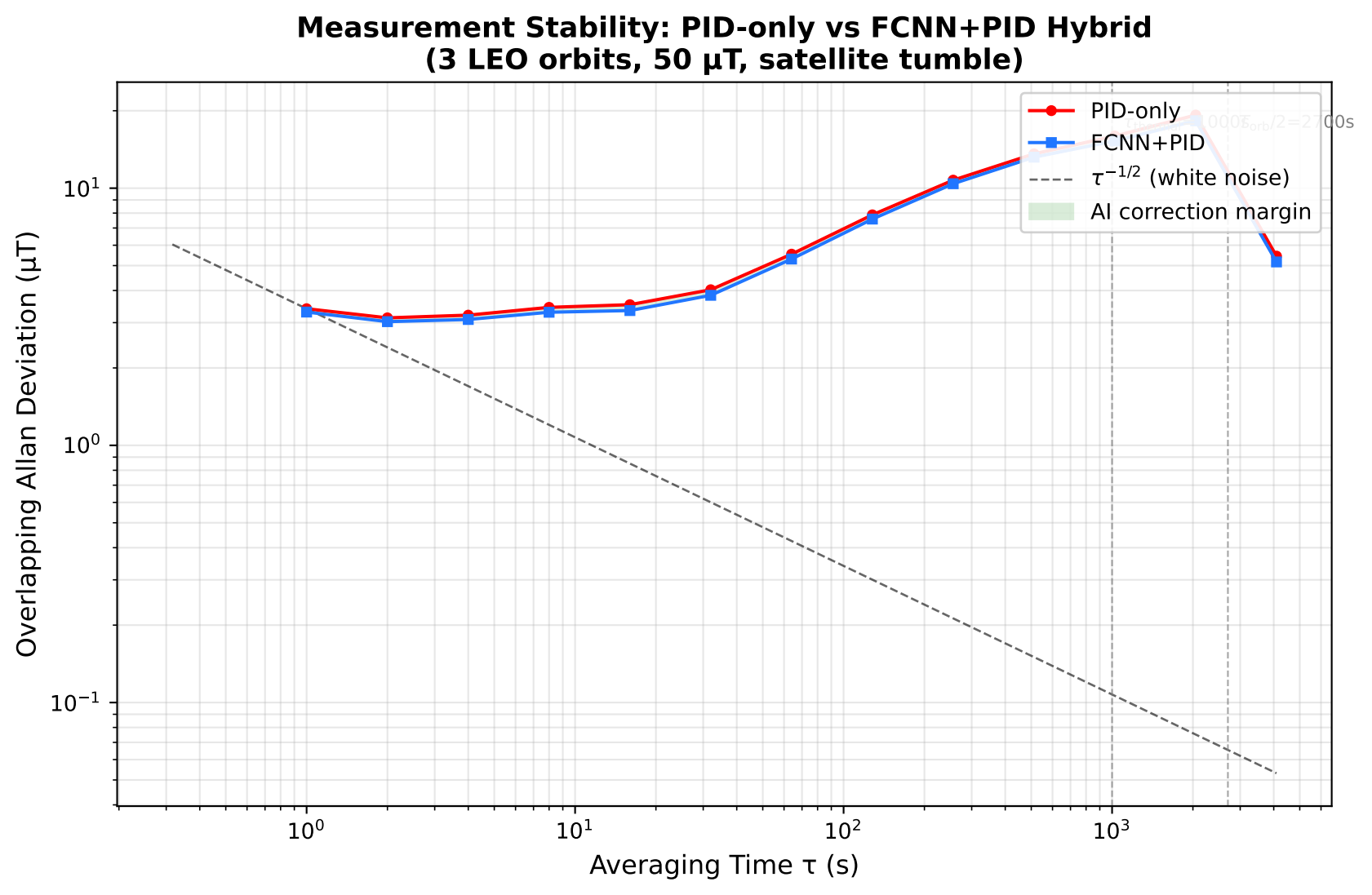
Measurement Stability: PID-only vs FCNN+PID Hybrid
(3 LEO orbits, 50 μT, satellite tumble)
PID-only
FCNN+PID
τ^−1/2 (white noise)
AI correction margin
Overlapping Allan Deviation (μT)
Averaging Time τ (s)
10^1
10^0
10^−1
10^0
10^1
10^2
10^3